

Lie's Theorem: A Structural Approach to Solvable Lie Algebras

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Abstract

This paper gives a clear and self-contained exposition of Lie's Theorem, a key result in the representation theory of solvable Lie algebras. The theorem states that any finite-dimensional representation of a solvable Lie algebra over an algebraically closed field of characteristic zero can be made upper triangular in a suitable basis. Following Serre's approach, the proof uses induction on the dimension of the representation space, with the existence of a common eigenvector as the main step. A key lemma on the behavior of characters over ideals supports the induction. Examples and corollaries illustrate the result and its link to solvability and simultaneous triangularization.

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1 Introduction

Lie's Theorem provides a striking linear-algebraic description of solvable Lie algebras: over an algebraically closed field of characteristic zero, every finite-dimensional representation can be realized by upper-triangular matrices. This means that solvable Lie algebras act in the simplest possible way—through successive one-dimensional invariant subspaces—so every irreducible representation is one-dimensional. The theorem thus reduces the representation theory of solvable algebras to scalar actions, revealing their structure in purely linear terms.

Beyond its intrinsic elegance, Lie's Theorem forms the foundation for deeper geometric results such as the Lie–Kolchin theorem, the structure of Borel subgroups, and the theory of flags in algebraic groups, where solvable subgroups correspond to upper-triangular forms. In this sense, Lie's Theorem serves as the conceptual bridge between abstract Lie algebra theory and the geometry of linear group actions.

2 Basics of Lie Algebras

2.1 Core Structure of Lie Algebras

Standing assumptions. Throughout, $\mathbb{k}$ is algebraically closed of characteristic 0 unless stated otherwise.

Definition 2.1. A *Lie algebra* over a field $\mathbb{k}$ is a vector space over $\mathbb{k}$ with a bilinear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ such that

1. $[x, y] = -[y, x]$ (antisymmetry),
2. $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$ (Jacobi identity),

for all $x, y, z \in \mathfrak{g}$.

Definition 2.2 (Lie subalgebra). A subspace $\mathfrak{h} \subset \mathfrak{g}$ is a *Lie subalgebra* if $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

2.2 Ideals, Quotients, and Representations

Definition 2.3 (Ideals and derived series). A subspace $\mathfrak{a} \subset \mathfrak{g}$ is an *ideal* (written $\mathfrak{a} \triangleleft \mathfrak{g}$) if $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$. The *derived algebra* is $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$, and the *derived series* is $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$. We say $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n .

Definition 2.4 (Representation and stable subspace). A *representation* of $\mathfrak{g}$ on a finite-dimensional space V is a Lie algebra homomorphism $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, i.e. $\rho([x, y]) = [\rho(x), \rho(y)]$. A subspace $W \subseteq V$ is $\mathfrak{g}$ -*stable* if $\rho(x)W \subseteq W$ for all $x \in \mathfrak{g}$.

Remark 2.5 (Stable line $\Leftrightarrow$ common eigenvector). A line $\mathbb{k}v \subset V$ is $\mathfrak{g}$ -stable iff there exists a linear form $\chi : \mathfrak{g} \rightarrow \mathbb{k}$ with $\rho(x)v = \chi(x)v$ for all $x \in \mathfrak{g}$ (i.e. v is a common eigenvector for $\rho(\mathfrak{g})$).

Definition 2.6 ($\mathfrak{g}$ -equivariant map). If (ρ_V, V) and (ρ_W, W) are $\mathfrak{g}$ -modules, a linear map $T : V \rightarrow W$ is $\mathfrak{g}$ -equivariant if $T(\rho_V(x)v) = \rho_W(x)T(v)$ for all $x \in \mathfrak{g}$, $v \in V$.

Proposition 2.7 (Quotient representation and equivariant quotient map). If $W \subset V$ is $\mathfrak{g}$ -stable for a representation (ρ, V) , then

$$\bar{\rho}(x)(v + W) := \rho(x)v + W \quad (x \in \mathfrak{g}, v \in V)$$

defines a representation $\bar{\rho} : \mathfrak{g} \rightarrow \mathfrak{gl}(V/W)$, and the quotient map $\pi : V \rightarrow V/W$, $\pi(v) = v + W$, is $\mathfrak{g}$ -equivariant: $\pi \circ \rho(x) = \bar{\rho}(x) \circ \pi$.

Proof. If $v + W = v' + W$ then $v - v' \in W$ and $\rho(x)(v - v') \in W$, so $\bar{\rho}$ is well defined. The homomorphism identity and the equivariance follow from those of ρ . $\square$

Definition 2.8 (Complete $\mathfrak{g}$ -stable flag). A chain $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ of $\mathfrak{g}$ -stable subspaces with $\dim(V_i/V_{i-1}) = 1$ is a *complete $\mathfrak{g}$ -stable flag*.

Definition 2.9 (Adjoint map and adjoint representation). For a Lie algebra $\mathfrak{g}$, the *adjoint map* of $X \in \mathfrak{g}$ is

$$\text{ad}_X : \mathfrak{g} \rightarrow \mathfrak{g}, \quad \text{ad}_X(Y) = [X, Y].$$

The *adjoint representation* is the Lie algebra homomorphism

$$\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g}), \quad X \mapsto \text{ad}_X,$$

characterized by $[\text{ad}_X, \text{ad}_Y] = \text{ad}_{[X, Y]}$.

Remark 2.10 (Ideals $\Leftrightarrow$ ad -stable subspaces). A subspace $I \subset \mathfrak{g}$ is an ideal iff it is $\text{ad}(\mathfrak{g})$ -stable:

$$I \text{ ideal} \iff [\mathfrak{g}, I] \subset I \iff \text{ad}_X(I) \subset I \text{ for all } X \in \mathfrak{g}.$$

3 Structural Tools for Lie's Theorem

Throughout this section $\mathbb{k}$ is algebraically closed of characteristic 0 and $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ denotes a finite-dimensional representation when relevant.

3.1 Solvability and Codimension-One Ideals

By definition 2.9 the derived series $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$ and $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$. A Lie algebra is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n .

Lemma 3.1. If $\mathfrak{g}$ is solvable and $\mathfrak{g} \neq 0$, then $\mathfrak{g}' \subsetneq \mathfrak{g}$.

Proof. Let $\mathfrak{g}^{(0)} = \mathfrak{g}$ and $\mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$. Since $\mathfrak{g}$ is solvable, choose $n \geq 1$ minimal with $\mathfrak{g}^{(n)} = 0$; then $\mathfrak{g}^{(n-1)} \neq 0$. If $\mathfrak{g}' = \mathfrak{g}$, then inductively $\mathfrak{g}^{(k)} = [\mathfrak{g}^{(k-1)}, \mathfrak{g}^{(k-1)}] = [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$ for all k , contradicting $\mathfrak{g}^{(n)} = 0$. Hence $\mathfrak{g}' \neq \mathfrak{g}$, and since always $\mathfrak{g}' \subseteq \mathfrak{g}$, we have $\mathfrak{g}' \subsetneq \mathfrak{g}$. $\square$

Proposition 3.2 (Existence of a codimension-one ideal). If $\mathfrak{g}$ is solvable and nonzero, then there exists a subspace $\mathfrak{h} \subset \mathfrak{g}$ such that

$$\mathfrak{g}' \subset \mathfrak{h}, \quad \dim(\mathfrak{g}/\mathfrak{h}) = 1, \quad \text{and} \quad \mathfrak{h} \triangleleft \mathfrak{g}.$$

Proof. By Lemma 3.1, $\mathfrak{g}' \subsetneq \mathfrak{g}$. Choose a vector space complement U of $\mathfrak{g}'$ in $\mathfrak{g}$, so $\mathfrak{g} = \mathfrak{g}' \oplus U$ with $U \neq 0$. Pick $u_0 \in U$, $u_0 \neq 0$, and define a linear functional

$$\ell : \mathfrak{g} \rightarrow \mathbb{k}, \quad \ell|_{\mathfrak{g}'} = 0, \quad \ell(u_0) = 1.$$

Then $\mathfrak{h} := \ker \ell$ is a hyperplane in $\mathfrak{g}$ with $\mathfrak{g}' \subset \mathfrak{h}$ and $\dim(\mathfrak{g}/\mathfrak{h}) = 1$.

$\mathfrak{h}$ is a Lie subalgebra. For $x, y \in \mathfrak{h}$, we have $[x, y] \in [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}' \subset \mathfrak{h}$, so $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$.

$\mathfrak{h}$ is an ideal. For $x \in \mathfrak{g}$, $y \in \mathfrak{h}$,

$$[x, y] \in [\mathfrak{g}, \mathfrak{h}] \subset [\mathfrak{g}, \ker \ell] \subset [\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}' \subset \mathfrak{h},$$

hence $[\mathfrak{g}, \mathfrak{h}] \subset \mathfrak{h}$. Thus $\mathfrak{h} \triangleleft \mathfrak{g}$, a codimension-one ideal containing $\mathfrak{g}'$. $\square$

4 Lie's Theorem

Throughout this section $\mathbb{k}$ is an algebraically closed field of characteristic 0. Let $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a finite-dimensional representation (Def. 2.11).

Proposition 4.1. *For a representation $\rho : \mathfrak{g} \rightarrow \mathfrak{gl}(V)$, the following are equivalent: (i) there exists a complete $\mathfrak{g}$ -stable flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ with $\dim(V_i/V_{i-1}) = 1$; (ii) there is a basis of V in which every $\rho(x)$ ($x \in \mathfrak{g}$) is upper triangular.*

proof sketch. (i) $\Rightarrow$ (ii): Choose a basis adapted to the flag; $\rho(x)V_i \subseteq V_i$ for all i , so each $\rho(x)$ is upper triangular. (ii) $\Rightarrow$ (i): In a basis making all $\rho(x)$ upper triangular, the standard flag $\langle e_1 \rangle \subset \langle e_1, e_2 \rangle \subset \cdots \subset \langle e_1, \dots, e_n \rangle$ is $\mathfrak{g}$ -stable with 1-dimensional steps. $\square$

Theorem 4.2 (Lie's theorem). *If $\mathfrak{g}$ is solvable (Def. 3.1), then there exists a complete $\mathfrak{g}$ -stable flag $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ with $\dim(V_i/V_{i-1}) = 1$. By Proposition 4.1, equivalently, there is a basis of V in which all $\rho(x)$, $x \in \mathfrak{g}$, are upper triangular.*

Reduction to a common eigenvector

It suffices to prove:

Theorem 4.3 (Reduction target). *Under the hypotheses of Theorem 4.2, there exist $0 \neq v \in V$ and a linear form $\chi : \mathfrak{g} \rightarrow \mathbb{k}$ such that $\rho(x)v = \chi(x)v$ for all $x \in \mathfrak{g}$.*

Proof that Theorem 4.3 $\Rightarrow$ Theorem 4.2. Assume $0 \neq v \in V$ and χ with $\rho(x)v = \chi(x)v$ for all x . Then $\mathbb{k}v$ is $\mathfrak{g}$ -stable, since $\rho(x)(\lambda v) = \lambda \chi(x)v \in \mathbb{k}v$. Let $\pi : V \rightarrow V/\mathbb{k}v$ be the quotient map. By Proposition 2.7 the quotient representation $\bar{\rho} : \mathfrak{g} \rightarrow \mathfrak{gl}(V/\mathbb{k}v)$ is well defined and π is $\mathfrak{g}$ -equivariant in the sense of Def. 2.16: $\pi \circ \rho(x) = \bar{\rho}(x) \circ \pi$.

Proceed by induction on $\dim V$. If $\dim V = 1$, the flag $0 \subset V$ works. For $\dim V > 1$, apply the induction hypothesis to $(\bar{\rho}, V/\mathbb{k}v)$ to get a complete $\mathfrak{g}$ -stable flag

$$0 = \bar{V}_0 \subset \bar{V}_1 \subset \cdots \subset \bar{V}_{n-1} = V/\mathbb{k}v, \quad \dim(\bar{V}_i/\bar{V}_{i-1}) = 1.$$

Pull back by π :

$$0 \subset V_1 := \mathbb{k}v \subset V_2 := \pi^{-1}(\bar{V}_1) \subset \cdots \subset V_n := \pi^{-1}(\bar{V}_{n-1}) = V.$$

Each V_i is $\mathfrak{g}$ -stable because π is $\mathfrak{g}$ -equivariant, and

$$V_i/V_{i-1} \cong \pi(V_i)/\pi(V_{i-1}) = \bar{V}_{i-1}/\bar{V}_{i-2}$$

is 1-dimensional. Hence this is a complete $\mathfrak{g}$ -stable flag. With respect to a basis adapted to the flag, all $\rho(x)$ are upper triangular, proving Theorem 4.2. $\square$

Lemma 4.4 (Main Lemma). *[1]*

Let $\mathfrak{h} \triangleleft \mathfrak{g}$ be an ideal and suppose there exist $v \neq 0$ and a linear form $\chi : \mathfrak{h} \rightarrow \mathbb{k}$ such that $\rho(h)v = \chi(h)v$ for all $h \in \mathfrak{h}$. Then

$$\chi([x, h]) = 0 \quad \text{for all } x \in \mathfrak{g}, h \in \mathfrak{h}.$$

Proof. Fix $x \in \mathfrak{g}$ and set $V_i := \text{span}\{v, \rho(x)v, \dots, \rho(x)^i v\}$ for $i \geq 0$. Choose n minimal with $V_n = V_{n+1}$. Note we can choose such n because V is finite-dimensional, the ascending chain $V_0 \subseteq V_1 \subseteq \dots$ stabilizes, so there exists a minimal n with $V_n = V_{n+1}$. Equivalently, $V_n = \text{span}\{v, \rho(x)v, \dots, \rho(x)^n v\}$ is the $\rho(x)$ -cyclic subspace generated by v .

Claim 1. For every $h \in \mathfrak{h}$ and $i \geq 0$,

$$\rho(h) \rho(x)^i v \equiv \chi(h) \rho(x)^i v \pmod{V_{i-1}}. \quad (1)$$

For $i = 0$ this is the hypothesis.

Assume (1) for $i - 1 \geq 0$. Since ρ is a Lie algebra homomorphism, we have

$$\rho([x, h]) = [\rho(x), \rho(h)] = \rho(x)\rho(h) - \rho(h)\rho(x).$$

Rearranging gives

$$\rho(h)\rho(x) = \rho(x)\rho(h) - \rho([x, h]).$$

Multiplying on the right by $\rho(x)^{i-1}v$ (with $i \geq 1$) yields

$$\begin{aligned} \rho(h)\rho(x)\rho(x)^{i-1}v &= (\rho(x)\rho(h) - \rho([x, h]))\rho(x)^{i-1}v \\ &= \rho(x)\rho(h)\rho(x)^{i-1}v - \rho([x, h])\rho(x)^{i-1}v \end{aligned}$$

For the *first term*, the induction hypothesis at level $i - 1$ says

$$\rho(h)\rho(x)^{i-1}v \equiv \chi(h)\rho(x)^{i-1}v \pmod{V_{i-2}}.$$

Applying $\rho(x)$ to both sides gives

$$\rho(x)\rho(h)\rho(x)^{i-1}v \equiv \chi(h)\rho(x)^i v \pmod{\rho(x)V_{i-2}}.$$

Since $\rho(x)V_{i-2} \subseteq V_{i-1}$ (multiplying each generator of V_{i-2} by $\rho(x)$ raises the power by 1), we obtain

$$\rho(x)\rho(h)\rho(x)^{i-1}v \equiv \chi(h)\rho(x)^i v \pmod{V_{i-1}}.$$

For the *second term*, note that $[x, h] \in \mathfrak{h}$, and (proved next) $\rho(\mathfrak{h})V_j \subseteq V_j$ for every j . In particular, with $j = i - 1$,

$$\rho([x, h])\rho(x)^{i-1}v \in V_{i-1},$$

so this term is 0 modulo V_{i-1} .

Combining the two parts yields

$$\rho(h)\rho(x)^i v \equiv \chi(h)\rho(x)^i v \pmod{V_{i-1}},$$

as claimed.

Claim 2. For every $j \geq 0$ and $h \in \mathfrak{h}$, $\rho(h)V_j \subseteq V_j$.

This is because, for a fixed $j \geq 0$, recall $V_j = \text{span}\{v, \rho(x)v, \dots, \rho(x)^j v\}$. By (1) with $i = j$ we have, for every $h \in \mathfrak{h}$,

$$\rho(h)\rho(x)^j v \equiv \chi(h)\rho(x)^j v \pmod{V_{j-1}},$$

i.e., there exists some $w \in V_{j-1}$ with

$$\rho(h) \rho(x)^j v = \chi(h) \rho(x)^j v + w \in \text{span}\{\rho(x)^j v\} + V_{j-1} \subseteq V_j.$$

Thus each generator $\rho(x)^j v$ of V_j is sent by $\rho(h)$ into V_j .

Now it's enough to check what $\rho(h)$ does to each generator $\rho(x)^j v$. By (1) (with $i = j$),

$$\rho(h) \rho(x)^j v = \chi(h) \rho(x)^j v + (\text{something in } V_{j-1}) \in V_j,$$

so $\rho(h)V_j \subseteq V_j$ (for $j = 0$ this is just $\rho(h)v = \chi(h)v \in V_0$).

Therefore, in the ordered basis $(v, \rho(x)v, \dots, \rho(x)^{n-1}v)$ of V_n , each $\rho(h)$ sends a basis vector to a scalar multiple of itself plus earlier ones. This means the matrix of $\rho(h)|_{V_n}$ is upper triangular, and all diagonal entries are $\chi(h)$. Hence

$$\text{Tr}_{V_n}(\rho(h)) = \underbrace{\chi(h) + \dots + \chi(h)}_{n \text{ times}} = n \chi(h).$$

Now replace h by $[x, h]$. Because $\rho([x, h]) = [\rho(x), \rho(h)]$ and V_n is invariant (by definition 2.12) under both $\rho(x)$ and $\rho(h)$, we can take traces on V_n and use the basic fact $\text{Tr}([A, B]) = 0$:

$$\text{Tr}_{V_n}(\rho([x, h])) = \text{Tr}_{V_n}([\rho(x), \rho(h)]) = 0.$$

The previous trace formula applied to $[x, h] \in \mathfrak{h}$ gives $\text{Tr}_{V_n}(\rho([x, h])) = n \chi([x, h])$. Therefore $n \chi([x, h]) = 0$, and since $n > 0$, we conclude $\chi([x, h]) = 0$.

Since $\rho(h)$ is linear and V_j is the span of its generators, it follows that $\rho(h)V_j \subseteq V_j$. $\square$

Proof of Theorem 4.3. We argue by induction on $m = \dim \mathfrak{g}$. If $m = 0$ there is nothing to prove.

Assume $m > 0$ and $\mathfrak{g}$ solvable. By solvability, $\mathfrak{g}' \subsetneq \mathfrak{g}$. By Proposition 3.2 there exists a codimension-one ideal $\mathfrak{h} \triangleleft \mathfrak{g}$ containing $\mathfrak{g}'$, hence $\mathfrak{h}$ is solvable.

By the induction hypothesis applied to $\mathfrak{h}$, there exist $v \neq 0$ and a character $\chi : \mathfrak{h} \rightarrow \mathbb{k}$ with $\rho(h)v = \chi(h)v$ for all $h \in \mathfrak{h}$. Let

$$W := \{w \in V \mid \rho(h)w = \chi(h)w \text{ for all } h \in \mathfrak{h}\}.$$

Then $0 \neq v \in W$. For $x \in \mathfrak{g}$, $h \in \mathfrak{h}$, $w \in W$ we have

$$\rho(h)(\rho(x)w) = \rho(x)\rho(h)w - \rho([x, h])w = \chi(h)\rho(x)w - \chi([x, h])w.$$

By Lemma 4.4, $\chi([x, h]) = 0$, hence $\rho(x)w \in W$. Thus W is $\mathfrak{g}$ -stable.

Choose $x \in \mathfrak{g} \setminus \mathfrak{h}$ (possible since $\text{codim } \mathfrak{h} = 1$). Because $\mathbb{k}$ is algebraically closed and $W \neq 0$ is $\rho(x)$ -stable, there exists $0 \neq v_0 \in W$ with $\rho(x)v_0 = \lambda v_0$ for some $\lambda \in \mathbb{k}$. By definition of W , $\rho(h)v_0 = \chi(h)v_0$ for all $h \in \mathfrak{h}$; since $\mathfrak{g} = \mathfrak{h} \oplus \mathbb{k}x$, the vector v_0 is a common eigenvector for $\rho(\mathfrak{g})$. $\square$

Proof of Theorem 4.2. By the first proof, Theorem 4.3 implies Theorem 4.2. Since Theorem 4.3 has now been proved, Lie's theorem follows. $\square$

4.1 Corollaries and discussion

Corollary 4.5. *If $\mathfrak{g}$ is solvable over an algebraically closed field of characteristic 0, then the adjoint representation $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$, $x \mapsto \text{ad}_x$ with $\text{ad}_x(y) = [x, y]$, admits a complete $\text{ad}(\mathfrak{g})$ -stable flag. Equivalently, there is a chain of ideals*

$$0 = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_n = \mathfrak{g}, \quad \dim(\mathfrak{g}_i/\mathfrak{g}_{i-1}) = 1.$$

Proof. By Definition 2.9, the adjoint representation $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ is given by $\text{ad}_X(Y) = [X, Y]$. For any $x, y \in \mathfrak{g}$ we have

$$[\text{ad}_x, \text{ad}_y] = \text{ad}_x \text{ad}_y - \text{ad}_y \text{ad}_x = \text{ad}_{[x, y]},$$

so $[\text{ad}(\mathfrak{g}), \text{ad}(\mathfrak{g})] = \text{ad}([\mathfrak{g}, \mathfrak{g}])$ and, by induction,

$$(\text{ad}(\mathfrak{g}))^{(k)} = \text{ad}(\mathfrak{g}^{(k)}).$$

If $\mathfrak{g}$ is solvable, then $\mathfrak{g}^{(m)} = 0$ for some m , hence $(\text{ad}(\mathfrak{g}))^{(m)} = \text{ad}(0) = 0$, so $\text{ad}(\mathfrak{g})$ is solvable. Applying Lie's theorem to the representation $\text{ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ yields a complete $\text{ad}(\mathfrak{g})$ -stable flag

$$0 = \mathfrak{g}_0 \subset \mathfrak{g}_1 \subset \cdots \subset \mathfrak{g}_n = \mathfrak{g}, \quad \dim(\mathfrak{g}_i/\mathfrak{g}_{i-1}) = 1.$$

By Remark 2.10, $\text{ad}(\mathfrak{g})$ -stability is equivalent to being an ideal:

$$\text{ad}_x(I) \subset I \ \forall x \in \mathfrak{g} \iff [\mathfrak{g}, I] \subset I \iff I \text{ is an ideal.}$$

Hence every $\mathfrak{g}_i$ in the flag is an ideal, proving the claim. $\square$

Corollary 4.6. *If $\mathfrak{g}$ is solvable as above, then $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.*

Proof. From Cor. 4.5, for all i and all $x \in \mathfrak{g}$, $y \in \mathfrak{g}_i$ we have $[x, y] \in \mathfrak{g}_{i-1}$, i.e. $[\mathfrak{g}, \mathfrak{g}_i] \subseteq \mathfrak{g}_{i-1}$. Let $\mathfrak{d} := [\mathfrak{g}, \mathfrak{g}]$ and define $\mathfrak{d}^{(1)} := \mathfrak{d}$, $\mathfrak{d}^{(k+1)} := [\mathfrak{d}, \mathfrak{d}^{(k)}]$. Inductively,

$$\mathfrak{d}^{(k)} \subseteq \mathfrak{g}_{n-k} \quad (1 \leq k \leq n),$$

so $\mathfrak{d}^{(n)} \subseteq \mathfrak{g}_0 = 0$. Hence $\mathfrak{d}$ is nilpotent. $\square$

Remark 4.7. The chain of ideals obtained in Corollary 4.5 is the Lie-algebraic precursor of the Lie-Kolchin theorem for connected solvable algebraic groups: such groups are simultaneously triangularizable in their finite-dimensional representations. In the setting of linear algebraic groups, these flags correspond to Borel subgroups and complete flags in flag varieties, showing that Lie's theorem captures, at the infinitesimal level, the geometric essence of solvable group actions.

5 Examples, field assumptions, and limitations

Recall that by Theorem 4.2, it demonstrates that over an algebraically closed field $\mathbb{k}$ of characteristic 0, a solvable Lie algebra $\mathfrak{g}$ acting on V admits a complete $\mathfrak{g}$ -stable flag, equivalently a basis in which all $\rho(x)$ are upper triangular. The examples below illustrate two key facts:

- (i) the theorem in action;
- (ii) why the field assumptions are necessary.

Example 5.1 (A positive example over $\mathbb{C}$). Work over $\mathbb{k} = \mathbb{C}$ and let $V = \mathbb{k}^2$ with basis (f_1, f_2) . Consider

$$X_f = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}, \quad Y_f = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}.$$

A direct computation gives

$$[Y_f, X_f] = Y_f X_f - X_f Y_f = X_f,$$

so $\mathfrak{g} = \text{span}_{\mathbb{k}}\{X_f, Y_f\}$ satisfies $[\mathfrak{g}, \mathfrak{g}] = \text{span}_{\mathbb{k}}\{X_f\}$ and $[\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}]] = 0$; hence $\mathfrak{g}$ is solvable.

Common eigenvector: Since $X_f^2 = 0$, X_f is nilpotent with only eigenvalue 0. For $v = (a, b)^T$,

$$X_f v = \begin{pmatrix} a+b \\ -a-b \end{pmatrix} = (a+b) \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad Y_f v = \begin{pmatrix} a \\ -a \end{pmatrix} = a \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence $\ker X_f = \text{span}_{\mathbb{k}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and the same vector $v_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ satisfies $X_f v_0 = 0$, $Y_f v_0 = v_0$.

Flag: Let $U = \text{span}_{\mathbb{k}}\{v_0\}$. Then U is $\mathfrak{g}$ -stable, and on V/U both X_f and Y_f act trivially. Thus the $\mathfrak{g}$ -stable flag $0 \subset U \subset V$ shows that all elements of $\mathfrak{g}$ act by upper triangular matrices—exactly as predicted by Lie’s theorem.

Why the hypotheses matter

We might wonder why the two points in the proof use them crucially:

(1) to extract a (common) eigenvector of $\rho(x)$ inside a nonzero invariant subspace we need eigenvalues to lie in the field—hence algebraic closure;

(2) the Main Lemma yields $n \chi([x, h]) = 0$ from a trace computation on an n -dimensional invariant subspace, and we may conclude $\chi([x, h]) = 0$ only when $\text{char } \mathbb{k} = 0$.

We will then show by counterexamples, If either assumption is dropped, Lie’s conclusion can fail.

Example 5.2 ($\mathbb{R}$ not algebraically closed). Let $\mathfrak{g} = \mathbb{R} \cdot x$ act on $V = \mathbb{R}^2$ by

$$\rho(x) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad (\text{rotation by } 90^\circ).$$

The characteristic polynomial is $\lambda^2 + 1$, so $\rho(x)$ has no real eigenvector. Over $\mathbb{C}$, however, the eigenvalues are $\pm i$ with eigenvectors $e_1 \pm ie_2$. Since $\mathfrak{g}$ is abelian (hence solvable) but V has no $\mathfrak{g}$ -stable line over $\mathbb{R}$, Lie’s theorem fails over $\mathbb{R}$ and holds after extending scalars to $\mathbb{C}$.

Example 5.3 (Positive characteristic: failure in char 2). Work over $\mathbb{k} = \mathbb{F}_2$ and let $V = \mathbb{k}^2$ with basis (e_1, e_2) . Set

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{and} \quad \mathfrak{g} = \text{span}_{\mathbb{k}}\{A, B\} \subset \mathfrak{gl}(V).$$

Since $\text{char } \mathbb{k} = 2$, the Lie bracket is $[A, B] = AB - BA = AB + BA$. A quick calculation gives

$$AB = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \Rightarrow [A, B] = AB + BA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = B.$$

Hence

$$\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] = \text{span}_{\mathbb{k}}\{B\}, \quad \mathfrak{g}^{(2)} = 0,$$

so $\mathfrak{g}$ is solvable.

The one-dimensional subspaces of V are $\mathbb{k}e_1$, $\mathbb{k}e_2$, and $\mathbb{k}(e_1 + e_2)$. Since $Be_1 = e_2$ and $Be_2 = e_1$, only $\mathbb{k}(e_1 + e_2)$ is B -invariant. However,

$$A(e_1 + e_2) = e_2 \notin \mathbb{k}(e_1 + e_2),$$

so no line is invariant under both A and B . Thus there is no $\mathfrak{g}$ -stable flag, and Lie's theorem fails in characteristic 2.

Proposition 5.4 (Failure of Lie's theorem in characteristic p). [2]

Let $\mathbb{k}$ be a field of characteristic $p > 0$, and $V = \mathbb{k}^p$. There exists a solvable Lie subalgebra $L \subset \mathfrak{gl}_p(\mathbb{k})$ for which V has no nonzero common eigenvector. Hence Lie's theorem fails in characteristic p .

Proof. Consider

$$x = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix}, \quad y = \text{diag}(0, 1, 2, \dots, p-1).$$

With $e_1, \dots, e_p$ the standard basis, $xe_i = e_{i+1}$ ($i < p$), $xe_p = e_1$, and $ye_i = (i-1)e_i$.

Commutator:

$$[y, x]e_i = ye_{i+1} - x((i-1)e_i) = e_{i+1} \quad (1 \leq i < p), \quad [y, x]e_p = ye_1 - x((p-1)e_p) = e_1,$$

so $[y, x] = x$, i.e. $[x, y] = x$. Hence

$$L = \text{span}_{\mathbb{k}}\{x, y\}, \quad L^{(1)} = [L, L] = \text{span}_{\mathbb{k}}\{x\}, \quad L^{(2)} = 0,$$

and L is solvable.

No common eigenvector: Assume $xv = \lambda(x)v$, $yv = \lambda(y)v$ for some $v = (v_1, \dots, v_p)^T \neq 0$. Then

$$xv = \begin{pmatrix} v_2 \\ v_3 \\ \vdots \\ v_p \\ v_1 \end{pmatrix} = \lambda(x) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_{p-1} \\ v_p \end{pmatrix}, \quad yv = \begin{pmatrix} 0 \\ 1v_2 \\ 2v_3 \\ \vdots \\ (p-1)v_p \end{pmatrix} = \lambda(y) \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_{p-1} \\ v_p \end{pmatrix}.$$

Comparing coordinates gives $\lambda(y)v_1 = 0$ and $((i-1) - \lambda(y))v_i = 0$ for $2 \leq i \leq p$, so $v = \alpha e_j$ with $\lambda(y) = j - 1$. Then

$$xv = x(\alpha e_j) = \begin{cases} \alpha e_{j+1}, & j < p, \\ \alpha e_1, & j = p, \end{cases}$$

If $j < p$: in coordinate $j + 1$, the left side is α , while the right side is 0, so $\alpha = 0$ (contradiction).

If $j = p$: in coordinate 1, the left side is α , the right side is 0, so $\alpha = 0$ (contradiction).

Thus no nonzero α works. Therefore the only simultaneous solution of

$$xv = \lambda(x)v, \quad yv = \lambda(y)v$$

is $v = 0$ which is a contradiction since v is non-zero. Thus Lie's theorem fails. $\square$

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6 appendix: Use of generate AI

Portions of this project were developed with the assistance of *ChatGPT* (OpenAI, 2025). The interaction scripts used for clarification of proofs and formatting are available at the following link: <https://chatgpt.com/share/69125d61-5a50-8000-8d17-e7cc04c76c0b>